

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Iris dataset
iris_data = load_iris()
df = pd.DataFrame(data=iris_data.data, columns=iris_data.feature_names)
df['species'] = iris_data.target
df['species_name'] = df['species'].map({0: 'setosa', 1: 'versicolor', 2: 'virginica'})

# Inspect dataset
df.head()

```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species	species_name
0	5.1	3.5	1.4	0.2	0	setosa
1	4.9	3.0	1.4	0.2	0	setosa
2	4.7	3.2	1.3	0.2	0	setosa
3	4.6	3.1	1.5	0.2	0	setosa
4	5.0	3.6	1.4	0.2	0	setosa

```

# Summary statistics and null check
print("Dataset Shape:", df.shape)
print("\nNull Values:\n", df.isnull().sum())
print("\nDescriptive Statistics:\n", df.describe())

# Visualizing feature distributions by species
sns.pairplot(df.drop(columns=['species']), hue='species_name')
plt.suptitle("Pairplot of Iris Features by Species", y=1.02)
plt.show()

# Boxplot of features
plt.figure(figsize=(10, 6))
sns.boxplot(data=df.drop(columns=['species', 'species_name']))
plt.title("Feature Distributions")
plt.show()

```

Dataset Shape: (150, 6)

Null Values:

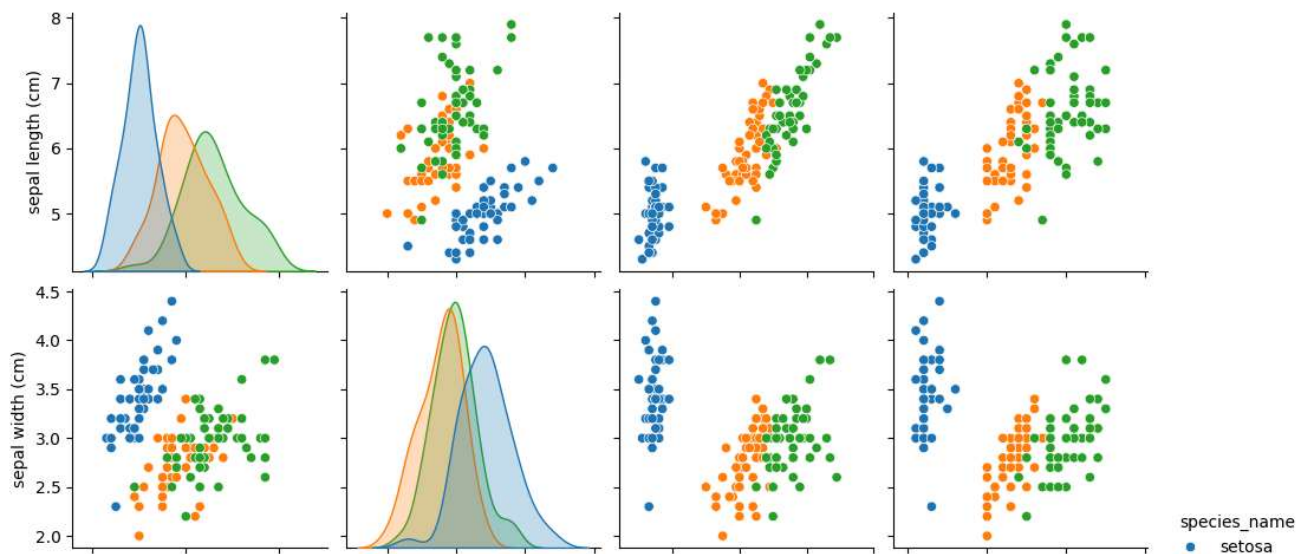
sepal length (cm) 0
sepal width (cm) 0
petal length (cm) 0
petal width (cm) 0
species 0
species_name 0
dtype: int64

Descriptive Statistics:

	sepal length (cm)	sepal width (cm)	petal length (cm) \
count	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000
std	0.828066	0.435866	1.765298
min	4.300000	2.000000	1.000000
25%	5.100000	2.800000	1.600000
50%	5.800000	3.000000	4.350000
75%	6.400000	3.300000	5.100000
max	7.900000	4.400000	6.900000

	petal width (cm)	species
count	150.000000	150.000000
mean	1.199333	1.000000
std	0.762238	0.819232
min	0.100000	0.000000
25%	0.300000	0.000000
50%	1.300000	1.000000
75%	1.800000	2.000000
max	2.500000	2.000000

Pairplot of Iris Features by Species



```
# Separate features and target
X = df[iris_data.feature_names]
y = df['species']
```

```
# 80/20 Train-Test split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
print(f"Training set size: {X_train.shape[0]}, Test set size: {X_test.shape[0]}")
```



```
# Initialize Classifiers
```

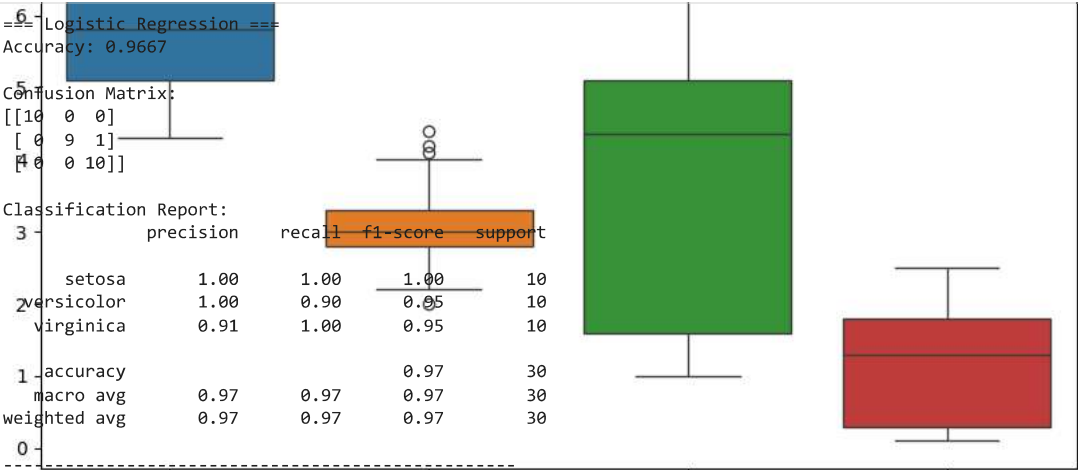
```
models = {
    "Logistic Regression": LogisticRegression(max_iter=200),
    "K-Nearest Neighbors": KNeighborsClassifier(n_neighbors=3)
}
```

```
# Train and evaluate models
```

```
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    print(f"=== {name} ===")
```

```
print(f"Accuracy: {acc:.4f}\n")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris_data.target_names))
print("-" * 50)
```



```
==== K-Nearest Neighbors ====
Accuracy: 1.0000

Confusion Matrix:
[[10  0  0]
 [ 0 10  0]
 [ 0  0 10]]

Classification Report:
precision    recall  f1-score   support

setosa      1.00     1.00     1.00        10
versicolor  1.00     1.00     1.00        10
virginica   1.00     1.00     1.00        10

accuracy    1.00     1.00     1.00        30
macro avg   1.00     1.00     1.00        30
weighted avg 1.00     1.00     1.00        30
```